

Lab 4: Web API CRUD Operation – Update Employee using HTTP PUT

Hands-on in this document

Objective

- To implement the HTTP PUT action method.
- To update employee details based on the user input.
- To validate employee ID before updating.
- To return appropriate responses using `ActionResult<Employee>`.
- To test the API using Swagger.

Procedure

Step 1: Open the Existing Web API Project

Opened the FirstWebAPI project in Visual Studio.

Step 2: Open EmployeeController

Opened the EmployeeController.cs file located inside the Controllers folder.

Step 3: Modify the PUT Action Method

Modified the PUT action method to return `ActionResult` instead of `IActionResult`.

Added validation to check whether the employee ID is valid.

Implemented the following validations:

- If the employee ID is less than or equal to zero, return

`BadRequest("Invalid employee id")`

- If the employee ID is not found in the hardcoded employee list, return

`BadRequest("Invalid employee id")`

- If the employee ID is valid, update the employee information using the JSON request body.
- Return the updated employee details using

`Ok(updatedEmployee)`

Step 4: Build the Project

Built the project successfully using

Build → Build Solution

No compilation errors were found.

Step 5: Run the Application

Executed the application using

F5

Swagger UI opened successfully.

Step 6: Authorize the Request

Clicked the Authorize button in Swagger.

Entered the Authorization header:

Bearer abc123

Clicked Authorize and then Close.

Step 7: Test the PUT Method

Opened

PUT /api/Employee/{id}

Clicked Try it Out.

Entered different employee IDs to verify the validations.

Test Case 1

Input

PUT /api/Employee/0

The image shows the Swagger UI interface for testing the PUT /api/Employee/{id} endpoint. The interface is divided into several sections:

- Method and Path:** PUT /api/Employee/{id}
- Parameters:** A table with columns 'Name' and 'Description'. It contains one parameter: 'id' (required), type 'integer(\$int32)', value '0', and location '(path)'. There are 'Cancel' and 'Reset' buttons.
- Request body:** A dropdown menu set to 'application/json'. Below it is a text area containing a JSON object:

```
{  "id": 0,  "name": "Akash",  "salary": 50000,  "permanent": true,  "department": {    "id": 101,    "name": "IT"  },  "skills": [    {      "id": 1,      "name": "C#"    }  ],  "dateOfBirth": "1999-05-10T00:00:00"}
```
- Buttons:** 'Execute' (blue) and 'Clear' (grey) buttons at the bottom.

Output

The image shows the Swagger UI interface displaying the server response for the PUT /api/Employee/0 endpoint. The interface is divided into several sections:

- Request URL:** https://localhost:7262/api/Employee/0
- Server response:** A table with columns 'Code' and 'Details'. It contains one response: '400' with details 'Error; response status is 400'.
- Response body:** A text area containing the response body: 'Invalid employee id'.
- Buttons:** 'Download' button at the bottom right.

Test Case 2

Input

PUT /api/Employee/10

The screenshot shows a REST client interface with the following details:

- Method:** PUT
- URL:** /api/Employee/{id}
- Parameters:** A table with one entry:

Name	Description
id * required Integer(Integer) (path)	10
- Request body:** application/json
- JSON Body:**

```
{  "id": 10,  "name": "Akash",  "salary": 50000,  "permanent": true,  "department": {    "id": 101,    "name": "IT"  },  "skills": [    {      "id": 1,      "name": "CS"    }  ],  "dateOfJoining": "1999-05-10T00:00:00"}
```
- Buttons:** Execute, Clear, Cancel, Reset

Output

The screenshot shows the server response details:

- Request URL:** https://localhost:7282/api/Employee/10
- Server response:**

Code	Details
400 Undocumented	Error: response status is 400 Response body Invalid employee id
- Buttons:** Download

Test Case 3

Input

PUT /api/Employee/1

PUT

/api/Employee/{id}

Parameters

Cancel

Reset

Name	Description
id	required ↑ Integer(Integer32) (path)

Request body

application/json

```
{  "id": 1,  "name": "Akash Kumar",  "salary": 65000,  "permanent": true,  "department": {    "id": 101,    "name": "IT"  },  "skills": [    {      "id": 1,      "name": "C#"    },    {      "id": 2,      "name": "ASP.NET Core"    }  ],  "dateOfBirth": "1999-05-10T00:00:00"}
```

Output

Request URL

https://localhost:7282/api/Employee/1

Server response

Code

Details

200

Response body

```
{  "id": 1,  "name": "Akash Kumar",  "salary": 65000,  "permanent": true,  "department": {    "id": 101,    "name": "IT"  },  "skills": [    {      "id": 1,      "name": "C#"    },    {      "id": 2,      "name": "ASP.NET Core"    }  ],  "dateOfBirth": "1999-05-10T00:00:00"}
```

Download